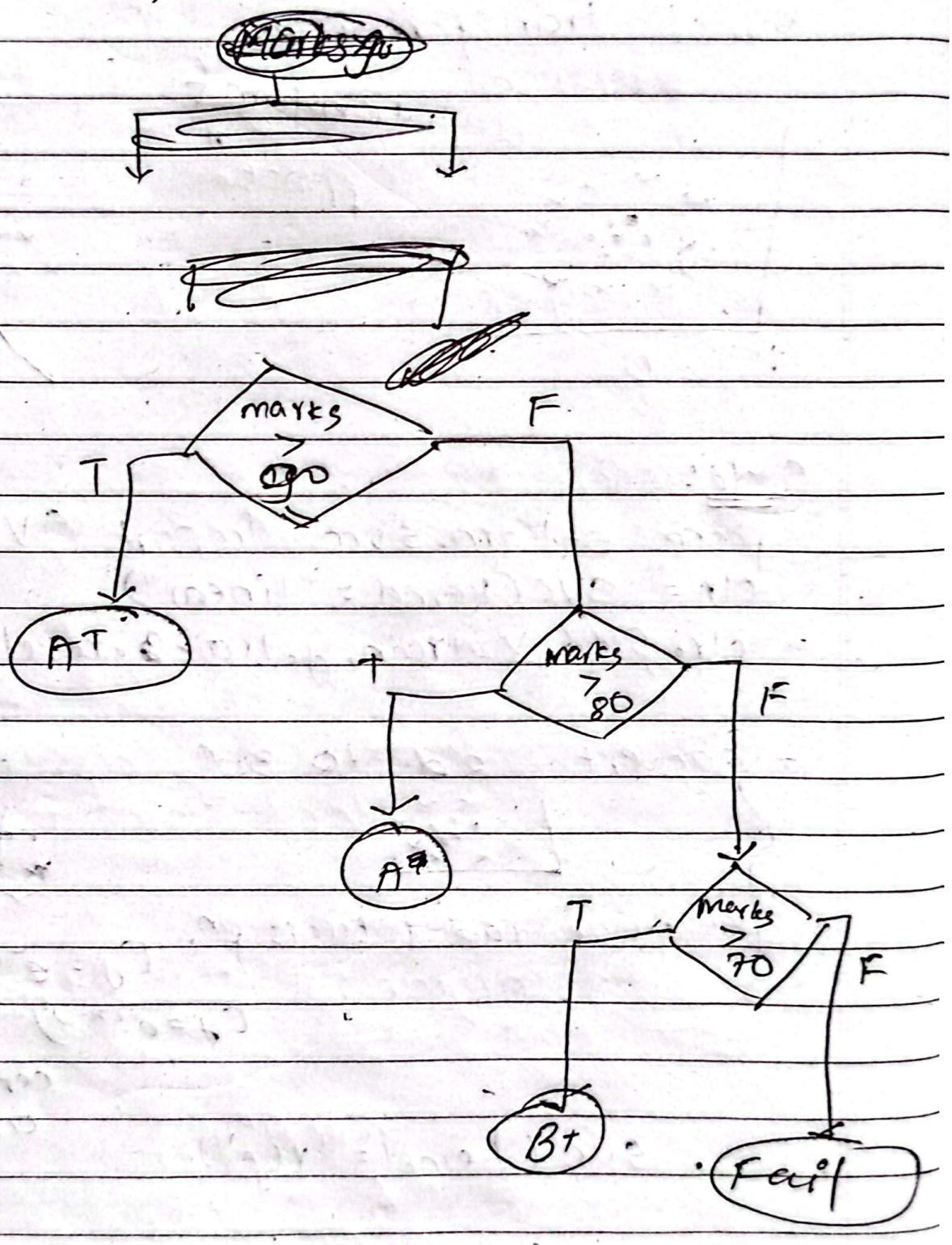
Decision Tree.

- Nested if else Statement.

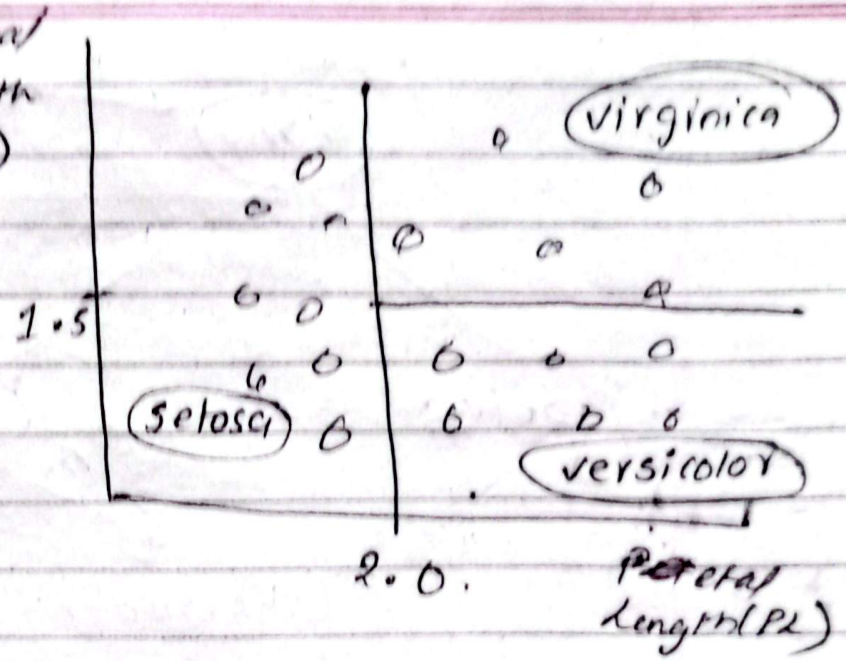


Page No.
Date: / /

Geometric intuition

root node:

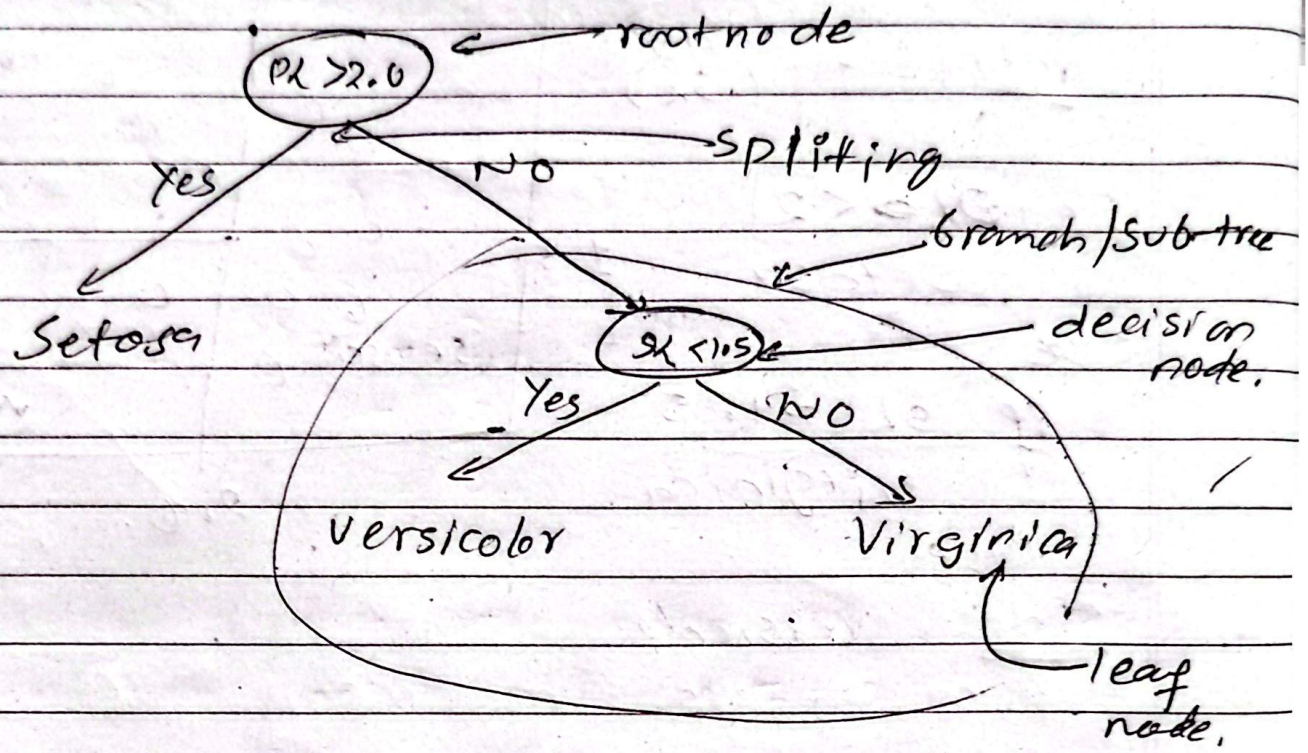
If. $(PL < 2)$
setosa
else
if SL > 1.5
virginica
else
versicolor.



Pseudo Code.
- determine the best feature [called root node]
- splitting based on root node
- Repeat.

~~Pattern~~ Pratically,
Decision tree → nested if-else statement

Mathematically,
Decision trees use hyperplanes which run parallel to anyone of the axes to cut your coordinate system into hyper cuboids.



Disadvantages
- Overfitting
- error in imbalance datasets.

CART - Classification and Regression Trees

Entropy:
- The measure of disorder.
- The measure of impurity ~~Impurity~~.